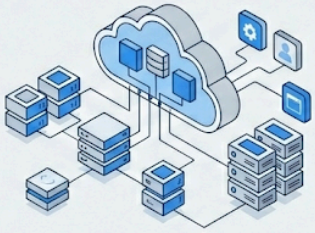
Kubernetes Interview Preparation Document



Enterprise Technical Documentation

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1. Introduction to Kubernetes

Kubernetes is an open-source container orchestration platform used to automate the deployment, scaling, and management of containerized applications. It helps organizations run applications in a highly available, scalable, and resilient environment. Kubernetes was originally developed by Google and is now maintained by the Cloud Native Computing Foundation (CNCF). In modern DevOps practices, Kubernetes plays a key role in enabling microservices architecture, automation, and cloud-native development.

Kubernetes provides features such as auto-scaling, self-healing, rolling updates, service discovery, and load balancing. It allows teams to efficiently manage application workloads across multiple environments, including on-premises, hybrid cloud, and public cloud platforms such as Azure Kubernetes Service (AKS), Amazon EKS, and Google GKE.

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2. Applications of Kubernetes

Kubernetes is widely used across industries to deploy and manage modern applications. It is commonly used for:

- Microservices-based application deployment.
- Continuous Integration and Continuous Delivery (CI/CD) automation.
- High-availability and fault-tolerant workloads.
- Cloud-native application modernization.
- AI/ML and data processing workloads.
- Hybrid and multi-cloud application deployments.

For example, organizations use Kubernetes to run scalable web applications, backend APIs, e-commerce platforms, and financial services workloads. In DevOps, it integrates with CI/CD tools like Azure DevOps, Jenkins, and GitHub Actions for automated deployments.

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3. Core Kubernetes Concepts

3.1 Cluster and Nodes

A Kubernetes cluster consists of a control plane and worker nodes. The control plane manages the cluster, while worker nodes run application workloads. The control plane includes components like API Server, Scheduler, Controller Manager, and etcd.

3.2 Pods

A Pod is the smallest deployable unit in Kubernetes. It contains one or more containers that share the same network and storage. Pods are ephemeral and replaced automatically in case of failure.

3.3 Deployments

Deployments manage stateless applications. They ensure desired state, enable rolling updates, and provide rollback capabilities.

3.4 Services

Services expose applications inside or outside the cluster. Types include ClusterIP, NodePort, and LoadBalancer.

3.5 ConfigMaps and Secrets

ConfigMaps store non-sensitive configuration data, while Secrets store sensitive information such as passwords and API keys.

3.6 Namespaces

Namespaces provide logical isolation and help manage multiple environments such as development, staging, and production.

3.7 Autoscaling

Horizontal Pod Autoscaler (HPA) scales applications based on CPU or memory usage.

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4. Kubernetes Integrations

Kubernetes integrates with many DevOps and cloud tools to build a complete automation ecosystem.

- CI/CD tools: Azure DevOps, Jenkins, GitHub Actions.
- Monitoring: Prometheus, Grafana, Azure Monitor.
- Logging: ELK stack, Fluentd.
- Security: HashiCorp Vault, Azure Key Vault.
- Service Mesh: Istio, Linkerd.
- Infrastructure as Code: Terraform and ARM templates.

In real-time projects, these integrations help automate deployments, improve observability, and enhance security.

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5. Real-World Experience

In my professional experience as a DevOps Engineer, I have used Kubernetes (AKS) to deploy containerized applications using Helm charts. I have designed CI/CD pipelines in Azure DevOps to automate build and deployment processes. I have worked on configuring namespaces, role-based access control (RBAC), resource limits, and autoscaling.

I also implemented monitoring using Prometheus and Grafana to track application performance. Additionally, I optimized infrastructure costs by right-sizing resources and configuring cluster autoscaler. I collaborated with developers and QA teams to troubleshoot deployment issues and ensure smooth releases.

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6. Scenario-Based Interview Questions and Answers

Q1. How would you handle a situation where your application pods are frequently restarting?

In such a scenario, I would first check pod logs using kubectl logs to identify the root cause. I would also inspect events using kubectl describe pod. Common reasons include memory limits, misconfiguration, or application errors. I would verify resource limits and adjust if needed. Additionally, I would check liveness and readiness probes and confirm they are properly configured.

Q2. How do you perform zero-downtime deployment in Kubernetes?

I use rolling updates with Deployments. Kubernetes gradually replaces old pods with new ones. I ensure readiness probes are configured so traffic is sent only to healthy pods. If any issue occurs, I perform a rollback using kubectl rollout undo.

Q3. How do you secure sensitive data in Kubernetes?

I store sensitive data in Secrets and integrate Kubernetes with external secret managers such as Azure Key Vault. I also implement RBAC and restrict access to namespaces.

Q4. How would you troubleshoot if a service is not accessible?

I would check service configuration, endpoints, and pod status. I verify whether the correct labels are used. I also check network policies, firewall rules, and ingress configuration.

Q5. How do you scale applications in Kubernetes?

I configure Horizontal Pod Autoscaler based on CPU or memory usage. For infrastructure scaling, I enable Cluster Autoscaler.

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7. Conclusion

Kubernetes is a critical skill for modern DevOps and cloud engineers. Strong knowledge of its architecture, troubleshooting, integrations, and real-time scenarios is essential for interviews. Practical experience with cloud platforms like AKS, automation, monitoring, and security helps demonstrate real-world expertise.

This document summarizes my Kubernetes preparation and real-world experience, which I can confidently explain in interviews.